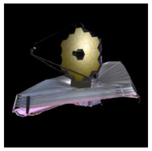

3245 - Up to the Task? A New Generation of Atmospheric and Interior Models of Brown Dwarfs for the JWST Era

Cycle: 2, Proposal Category: AR

INVESTIGATORS

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OBSERVATIONS

ABSTRACT

JWST's spectroscopic capabilities, along with its ability to measure brown dwarf luminosities with unprecedented precision, will bring about revolutionary advances to brown dwarf science. However, theoretical atmospheric and evolutionary models are critical to enable these advances. JWST data has already shown that available state-of-the-art models for brown dwarfs are insufficient to leverage their full scientific potential. We aim to upgrade these models in three major ways. First, both clouds and vertical mixing are expected to significantly influence the atmospheres of JWST targets like early T-dwarfs (with silicate clouds) and Y-dwarfs (water clouds). However, no atmospheric model has sufficiently explored clouds and vertical mixing simultaneously and self-consistently. We propose to upgrade our atmospheric models to treat both clouds and vertical mixing in a self-consistent framework. Second, we propose to build a grid of brown dwarf atmospheric models, with a large range of previously under-explored parameters like metallicity, atmospheric C/O ratio, and atmospheric vertical mixing. Such a grid will be important to constrain the atmospheric chemistry of brown dwarfs from JWST spectroscopy. Lastly, to properly connect the very precise JWST luminosity measurements to brown dwarf mass and radius, we require evolutionary models that are sensitive to parameters like metallicity, C/O ratio, and atmospheric mixing.

Such models are not yet available. We will compute a new generation of evolutionary models for brown dwarfs, sensitive to these parameters, which will enable the community to reliably connect the bulk properties of brown dwarfs to their measured luminosities.

OBSERVING DESCRIPTION

Not Applicable.